Big Mountain Resort, Montana: Ticket Price Data Analysis

Following the installation of an additional chair and associated increase in the operating cost of $1.54M, we have performed competitive market analysis to determine the optimal ticket price. Additionally, we have evaluated several proposed scenarios for future expansion and downsizing with impact on ticket prices and revenue in mind. The following are our recommendations as well as breakdown of the methodology that led us to them.

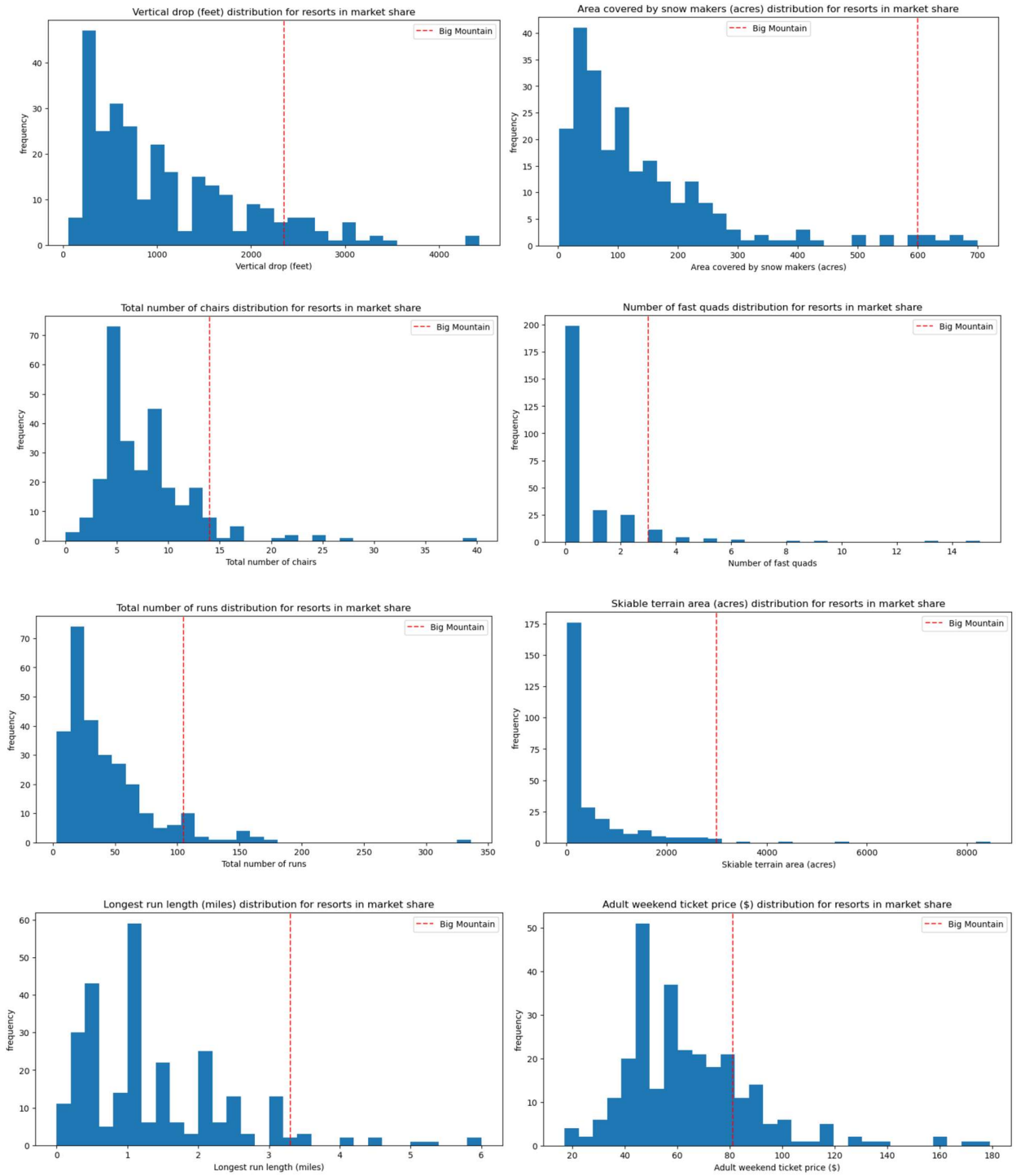
Recommendation: we recommend increasing ticket prices to at least $94 / person to be on par with the average price of resorts in Big Mountain Resort's market segment. This would result in additional annual revenue of $22.75M (assuming 350k visitors purchasing ~5 lift tickets each). The second of the proposed scenarios: "Increase the vertical drop by adding a run to a point 150 feet lower down but requiring the installation of an additional chair lift to bring skiers back up, without additional snow making coverage" would warrant an additional price increase to $95 / person resulting in extra annual revenue of $1.95M. Additional analysis of a narrower slice of US resorts indicates that price increase to $100-105 / person range would be a premium charge above the average price of resorts in BMR's market segment and would capitalize on BMR's outstanding terrain and facilities.

Methodology: Starting with a database of US ski resorts that includes various stats as well as ticket prices, we have cleaned the data and performed exploratory data analysis to better understand what aspects of terrain, location, size, etc. have the biggest impact on ticket prices. We have zeroed in on resorts' weekend prices, determined that the state where the resort is located has little impact on ticket prices and excluded a few resorts from this study that are missing weekend price data. Adding state sizes and populations to the data base allowed us to determine various ratios showing scarcity of resorts in their state, the proportion of night skiing a particular resort has in its state, etc. This ultimately, didn't have any impact on our findings.

Following that we moved forward with pre-processing and training, eventually finding optimal models that could predict ticket prices with highest accuracy given all the resort's stats. Scikit-learn (sklearn) built on Python 3.0 was used to train and optimize ML models. Linear vs random forest models were evaluated with the latter giving slightly more consistent results.

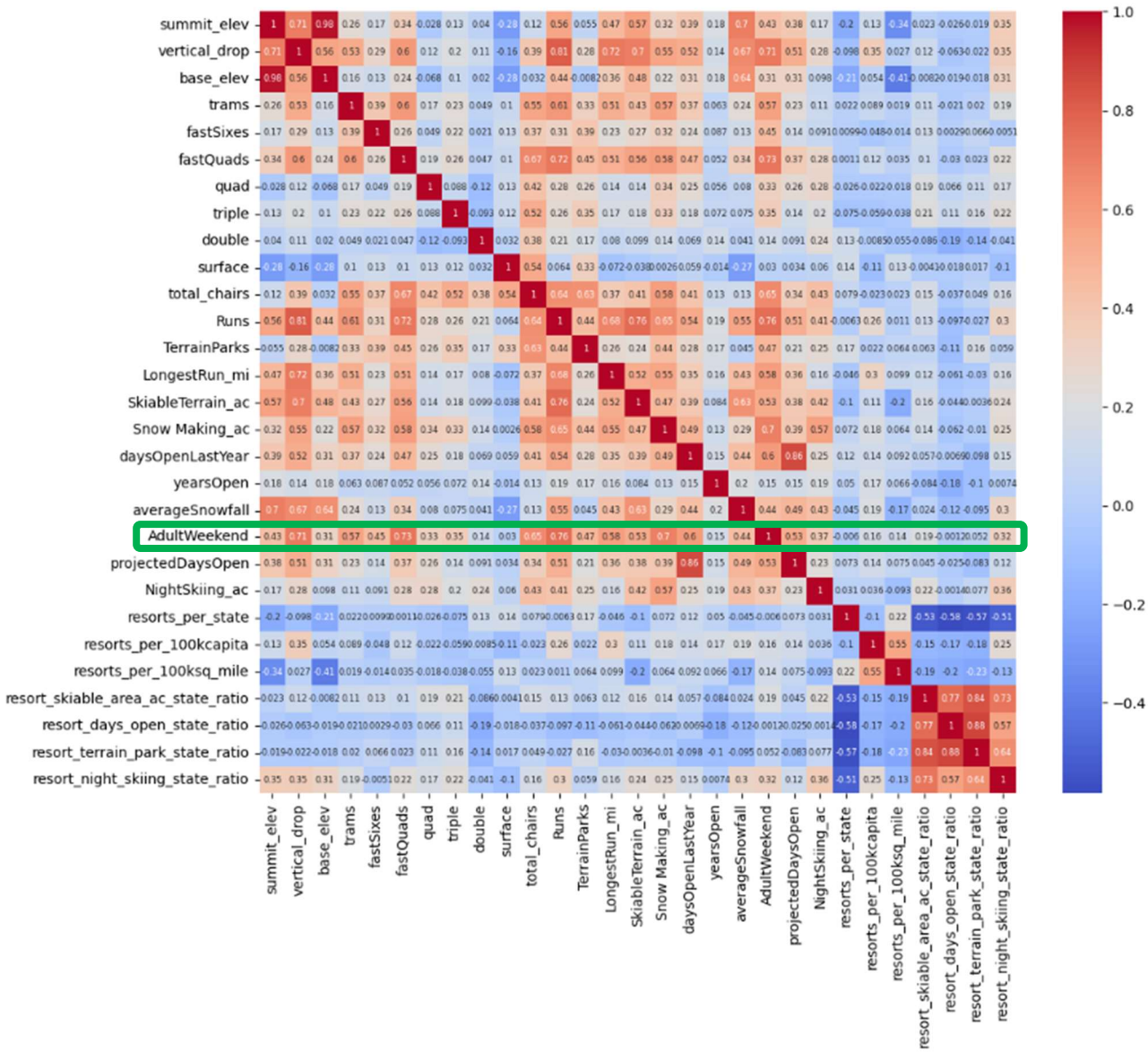
Findings: Data analysis of all US ski resorts revealed several metrics that have an outsized influence on ticket prices. Maximum vertical drop, area covered by snow making, total number of runs and chairs and more specifically fast quads, skiable terrain area, the length of the longest runs and number of days open per year all have strong positive correlations ($p > 0.5$) with ticket prices (appendix). Below charts show BMR's relative position in the resort

breakdown by stats. While BMR has outstanding terrain features and facilities, it is only slightly above average in terms of lift ticket prices.



Future work: Future model would ideally include information such as number of annual visitors and whether ticket prices are over- or underpriced. Proximity to large population areas or airports was not considered in our model at this time. It is not clear from the our

study if "uncrowded" resort could be of higher value to the customers or the inconvenience of a full day of travel would cancel out that advantage.

With more data BMR could make this model more useful and include a more dynamic pricing model that factors in holidays and specific months of high demand/high snow accumulation to be priced higher as a lot of other major resorts do. We believe more work could be done to focus on improving the aspects of the skiing experience that are more valuable to the skiers and increase the revenue for the BMR.

Appendix: Heatmap of correlations in the ski data for the US resorts.